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Learning Objectives:

Upon completion of this activity, participants will be able to:

- Integrate into professional practice the updates to the NCCN Guidelines for Colorectal Cancer Screening
- Describe the rationale behind the decision-making process for developing the NCCN Guidelines for Colorectal Cancer Screening

Disclosure of Relevant Financial Relationships

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Individuals Who Provided Content Development and/or Authorship Assistance:

The faculty listed below have no relevant financial relationship(s) with ineligible companies to disclose.
Xavier Llor, MD, PhD, Panel Vice Chair
Surya Gurudu, MD, Panel Member
Jennifer Maratt, MD, MS, Panel Member
Arnold J. Markowitz, MD, Panel Member
Carly J. Cassara, MSc, Guidelines Layout Specialist, NCCN
Vaishnavi Sambandam, PhD, Oncology Scientist/Medical Writer, NCCN

The faculty listed below have the following relevant financial relationship(s) with ineligible companies to disclose. All of the relevant financial relationships listed for these individuals have been mitigated.

Reid M. Ness, MD, MPH, Panel Chair, has disclosed receiving grant/research support from Guardant Health.
Mohammad Ali Abbass, MD, MPH, Panel Member, has disclosed receiving consulting fees from Invitae.

To view disclosures of external relationships for the NCCN Guidelines panel, go to NCCN.org/guidelines/guidelines-panels-and-disclosure/disclosure-panels

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Colorectal Cancer Screening, Version 1.2024
Featured Updates to the NCCN Guidelines

Reid M. Ness, MD, MPH[1,*]; Xavier Llor, MD, PhD[2,*]; Mohammad Ali Abbass, MD, MPH[3,*]; Shrinivas Bishu, MD[4]; Christopher T. Chen, MD[5]; Gregory Cooper, MD[6]; Dayna S. Early, MD[7]; Mark Friedman, MD[8]; David Fudman, MD[9]; Francis M. Giardiello, MD, MBA[10]; Kathryn Glaser, PhD, MA[11]; Surya Gurudu, MD[12,*]; Michael Hall, MD, MS[13]; Lyen C. Huang, MD, MPH[14]; Rachel Issaka, MD, MAS[15]; Bryson Katona, MD, PhD[16]; Trilokesh Kidambi, MD[17]; Audrey J. Lazenby, MD[18]; Jennifer Maratt, MD, MS[19,*]; Arnold J. Markowitz, MD[20,*]; Joseph Marsano, MD[21]; Folasade P. May, MD, PhD, MPhil[22]; Robert J. Mayer, MD[23]; Kinga Olortegui, MD[24]; Swati Patel, MD, MS[25]; Shajan Peter, MD[26]; Laura D. Porter, MD[27]; Mehnaz Shafi, MD[28]; Peter P. Stanich, MD[29]; Jonathan Terdiman, MD[30]; Peter Vu, MD[31]; Jennifer M. Weiss, MD, MS[32]; Elizabeth Wood, MD[33]; Carly J. Cassara, MSc[34,*]; and Vaishnavi Sambandam, PhD[34,*]

Abstract

The NCCN Guidelines for Colorectal Cancer (CRC) Screening describe various colorectal screening modalities as well as recommended screening schedules for patients at average or increased risk of developing sporadic CRC. They are intended to aid physicians with clinical decision-making regarding CRC screening for patients without defined genetic syndromes. These NCCN Guidelines Insights focus on select recent updates to the NCCN Guidelines, including a section on primary and secondary CRC prevention, and provide context for the panel's recommendations regarding the age at which to initiate screening in average-risk individuals and those with increased risk based on personal history of childhood, adolescent, and young adult cancer.

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Overview

Colorectal cancer (CRC) is the third most commonly diagnosed cancer and the second most common cause of cancer death in the United States.[1] It is also the leading cause of cancer death in men and second in women aged <50 years in the United States. In 2024, an estimated 106,590 new cases of colon cancer and 46,220 new cases of rectal cancer will occur in the United States.[2] During the same year, it is estimated that 52,010 people will die of colon and rectal cancer.[2] Screening individuals for CRC can reduce mortality by detecting cancer at an early, curable stage and may decrease incidence by detecting and removing precancerous polyps.[3,4]

Patients with localized CRC have a 91% relative 5-year survival rate, whereas rates for those with regional and distant disease are 73% and 14%, respectively, demonstrating that earlier diagnosis can have a large impact on survival.[1] The incidence of CRC continued to trend downward from 54.5 to 38.6 per 100,000 people from 2000 to 2014[5] and to 35.6 from 2015 to 2019, whereas the 5-year mortality rate from CRC was 45% between 2011 and 2015 and decreased to 13.2% between 2016 and 2020.[1] These improvements in the incidence of and mortality from CRC over past years are thought, at least in part, to be a result of cancer prevention and earlier diagnosis through screening and better treatment modalities.[2] According to the CDC, the screening rate in the United States among adults aged 50 to 75 years increased from approximately 42% in 2000 to 72.2% in 2021.[6–8] Conversely, the incidence rates of colon and rectal cancers in adults aged <50 years have been increasing by approximately 2% per year since 2003.[1,2]

In general, CRC incidence rates have doubled in younger adults, and most CRC cases in adolescent and young adult (AYA) individuals appear to be sporadic.[9] Causes for this increase in early-onset CRC are unknown and may be attributable to diet, environmental, and lifestyle factors. By 2030, it is estimated that 15% of CRCs will be diagnosed in younger adults, and therefore screening strategies need to be updated for appropriate screening and timely diagnosis in this population.[10]

These NCCN Guidelines Insights focus on recent changes in the NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines) for CRC Screening, including a summary of relevant

[1]Vanderbilt-Ingram Cancer Center; [2]Yale Cancer Center/Smilow Cancer Hospital; [3]Robert H. Lurie Comprehensive Cancer Center of Northwestern University; [4]University of Michigan Rogel Cancer Center; [5]Stanford Cancer Institute; [6]Case Comprehensive Cancer Center/University Hospitals Seidman Cancer Center and Cleveland Clinic Taussig Cancer Institute; [7]Siteman Cancer Center at Barnes-Jewish Hospital and Washington University School of Medicine; [8]Moffitt Cancer Center; [9]UT Southwestern Simmons Comprehensive Cancer Center; [10]Johns Hopkins Kimmel Cancer Center; [11]Roswell Park Comprehensive Cancer Center; [12]Mayo Clinic Comprehensive Cancer Center; [13]Fox Chase Cancer Center; [14]Huntsman Cancer Institute at the University of Utah; [15]Fred Hutchinson Cancer Center; [16]Abramson Cancer Center at the University of Pennsylvania; [17]City of Hope National Medical Center; [18]Fred & Pamela Buffett Cancer Center; [19]Indiana University Melvin and Bren Simon Comprehensive Cancer Center; [20]Memorial Sloan Kettering Cancer Center; [21]UC Davis Comprehensive Cancer Center; [22]UCLA Jonsson Comprehensive Cancer Center; [23]Dana-Farber Cancer Institute; [24]The UChicago Medicine Comprehensive Cancer Center; [25]University of Colorado Cancer Center; [26]O'Neal Comprehensive Cancer Center at UAB; [27]Independent Patient Advocate; [28]The University of Texas MD Anderson Cancer Center; [29]The Ohio State University Comprehensive Cancer Center - James Cancer Hospital and Solove Research Institute; [30]UCSF Helen Diller Family Comprehensive Cancer Center; [31]UC San Diego Moores Cancer Center; [32]University of Wisconsin Carbone Cancer Center; [33]The University of Tennessee Health Science Center; and [34]National Comprehensive Cancer Network.

*Provided content development and/or authorship assistance.

The full and most current version of these NCCN Guidelines is available at NCCN.org.

NCCN CATEGORIES OF EVIDENCE AND CONSENSUS

Category 1: Based upon high-level evidence, there is uniform NCCN consensus that the intervention is appropriate.
Category 2A: Based upon lower-level evidence, there is uniform NCCN consensus that the intervention is appropriate.
Category 2B: Based upon lower-level evidence, there is NCCN consensus that the intervention is appropriate.
Category 3: Based upon any level of evidence, there is major NCCN disagreement that the intervention is appropriate.

All recommendations are category 2A unless otherwise indicated.

NCCN CATEGORIES OF PREFERENCE

Preferred intervention: Interventions that are based on superior efficacy, safety, and evidence; and, when appropriate, affordability.
Other recommended intervention: Other interventions that may be somewhat less efficacious, more toxic, or based on less mature data; or significantly less affordable for similar outcomes.
Useful in certain circumstances: Other interventions that may be used for selected patient populations (defined with recommendation).

All recommendations are considered appropriate.

Clinical trials: NCCN believes that the best management for any patient with cancer is in a clinical trial. Participation in clinical trials is especially encouraged.

PLEASE NOTE

The NCCN Guidelines® are a statement of evidence and consensus of the authors regarding their views of currently accepted approaches to treatment.

The NCCN Guidelines® Insights highlight important changes in the NCCN Guidelines® recommendations from previous versions. Colored markings in the algorithm show changes and the discussion aims to further understanding of these changes by summarizing salient portions of the panel's discussion, including the literature reviewed.

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data supporting the recent updates to dietary factors and aspirin recommendations in the primary and secondary CRC prevention section, and discussions related to the age at which to initiate screening in average-risk individuals and those with increased risk based on personal history of childhood, adolescent, and young adult cancer, with concluding statements on the specific changes made to the NCCN Guidelines.

Primary and Secondary Prevention of CRC

There are several nonmodifiable risk factors, including age and hereditary factors, and modifiable risk factors, including environmental and lifestyle factors, that are associated with the occurrence of CRC (Figure 1).[11] Epidemiologic studies have identified several environmental factors (eg, dietary) and pharmacologic factors that are associated with CRC risk, and these factors are an important adjunct to the primary and secondary prevention of CRC.[12]

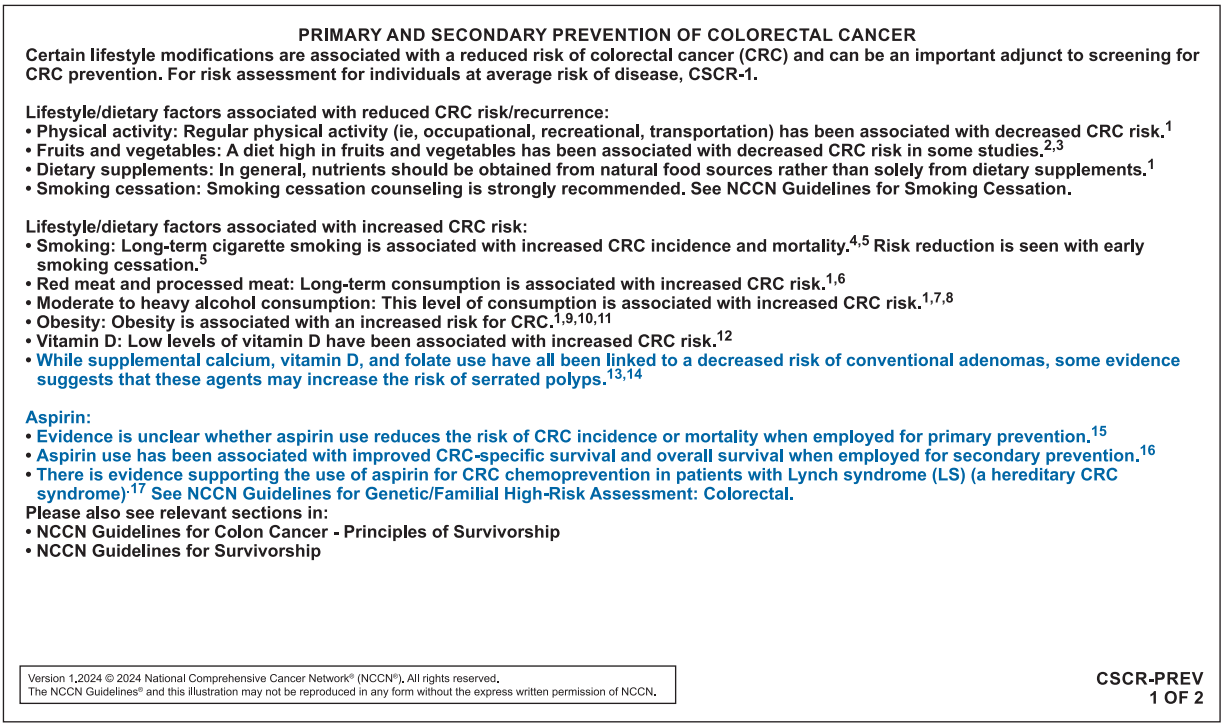
Dietary Factors (Vitamin D, Calcium, and Folate)

Increasing evidence supports that serrated polyps (SPs) represent another precursor lesion of CRC and contribute to approximately one-third of CRC cases through an alternative pathway.[13] The serrated pathway plays an important role in the development of "interval cancers," which occur despite appropriately timed endoscopic surveillance. In contrast to the conventional pathway arising from chromosomal instability, the serrated pathway is characterized by the CpG island methylation phenotype, *BRAF* mutation, and often microsatellite instability.[14,15]

Among lifestyle factors, smoking has been differentially strongly associated with SPs more than conventional adenomas.[16,17] However, for other factors, epidemiologic data are sparse and inconsistent. He et al[18] performed a comprehensive analysis of the risk factor profiles of SPs and conventional adenomas within 3 large prospective cohort studies—the Nurses' Health Study (NHS), the NHS2, and the Health Professionals Follow-up Study (HPFS)—and assessed several CRC risk factors with respect to SPs and conventional adenomas. The analysis comprised 141,143 participants who had undergone lower gastrointestinal endoscopy, provided updated diet and lifestyle data every 2 to 4 years, and were followed until diagnosis of a first polyp. Thirteen risk factors for CRC were assessed in patients with SPs or conventional adenomas, and their associations with histopathology features were also examined.

During the 18 to 20 years of follow-up, 7,945 SPs, 9,212 conventional adenomas, and 2,382 synchronous SPs and conventional adenomas were documented.[18] Although lifestyle factors such as smoking and alcohol intake were associated with higher risk of SPs and conventional adenomas, higher intake of vitamin D was associated with lower risk, and inversely associated with SPs (odds ratio [OR], 0.92; 95% CI, 0.86–0.98) and conventional adenomas (OR, 0.85; 95% CI, 0.80–0.90). Total folate and calcium intake were also inversely associated with risk of conventional adenomas (OR, 0.93 [95% CI, 0.87–0.99] and 0.90 [95% CI, 0.85–0.96], respectively). Although this study and others[19–21] favor the association of vitamin D, calcium, and folate with reduced CRC risk, there are several other conflicting studies that highlight the potential harmful effects of these supplements in increasing the risk of CRC.[22,23]

In a randomized, multicenter, double-blind, placebo-controlled study that aimed to determine the effects of calcium and vitamin D supplementation on the incidence of sessile serrated lesions (SSLs; also known as *sessile serrated polyps* or *sessile serrated adenomas* [SSP/SSAs]), Crockett et al[24] found evidence that calcium and vitamin D supplementation increased the risk of SSLs. This was noted to be a late effect, occurring 6 to 10 years after supplementation began. In this trial, patients between ages 45 and 75 years who had a clearing colonoscopy and a scheduled surveillance colonoscopy in 3 or 5 years were invited to participate. Among 2,813 eligible participants, 2,259 were randomized to 1 of 4 treatment groups using a partial 2×2 factorial design: 419 to the calcium carbonate group (1,200 mg elemental calcium/day), 420 to vitamin D3 group (1,000 IU/day), 421 to both agents, and 415 to the placebo group. While in the treatment phase, there was no effect of either calcium or vitamin D on incidence of SSLs, whereas during the later observational phase, elevated risks of SSLs were found to be associated with calcium alone and calcium + vitamin D treatment (adjusted risk ratio [RR], 2.66 [95% CI, 1.44–4.89] and 3.82 [95% CI, 1.26–11.57], respectively). This



PRIMARY AND SECONDARY PREVENTION OF COLORECTAL CANCER

Certain lifestyle modifications are associated with a reduced risk of colorectal cancer (CRC) and can be an important adjunct to screening for CRC prevention. For risk assessment for individuals at average risk of disease, CSCR-1.

Lifestyle/dietary factors associated with reduced CRC risk/recurrence:
- Physical activity: Regular physical activity (ie, occupational, recreational, transportation) has been associated with decreased CRC risk.[1]
- Fruits and vegetables: A diet high in fruits and vegetables has been associated with decreased CRC risk in some studies.[2,3]
- Dietary supplements: In general, nutrients should be obtained from natural food sources rather than solely from dietary supplements.[1]
- Smoking cessation: Smoking cessation counseling is strongly recommended. See NCCN Guidelines for Smoking Cessation.

Lifestyle/dietary factors associated with increased CRC risk:
- Smoking: Long-term cigarette smoking is associated with increased CRC incidence and mortality.[4,5] Risk reduction is seen with early smoking cessation.[5]
- Red meat and processed meat: Long-term consumption is associated with increased CRC risk.[1,6]
- Moderate to heavy alcohol consumption: This level of consumption is associated with increased CRC risk.[1,7,8]
- Obesity: Obesity is associated with an increased risk for CRC.[1,9,10,11]
- Vitamin D: Low levels of vitamin D have been associated with increased CRC risk.[12]
- While supplemental calcium, vitamin D, and folate use have all been linked to a decreased risk of conventional adenomas, some evidence suggests that these agents may increase the risk of serrated polyps.[13,14]

Aspirin:
- Evidence is unclear whether aspirin use reduces the risk of CRC incidence or mortality when employed for primary prevention.[15]
- Aspirin use has been associated with improved CRC-specific survival and overall survival when employed for secondary prevention.[16]
- There is evidence supporting the use of aspirin for CRC chemoprevention in patients with Lynch syndrome (LS) (a hereditary CRC syndrome).[17] See NCCN Guidelines for Genetic/Familial High-Risk Assessment: Colorectal.

Please also see relevant sections in:
- NCCN Guidelines for Colon Cancer - Principles of Survivorship
- NCCN Guidelines for Survivorship

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CSCR-PREV
1 OF 2

Figure 1. CSCR-PREV 1 of 2. NCCN Clinical Practice Guidelines in Oncology for Colorectal Cancer Screening, Version 1.2024.

study highlighted the increased risk of SSLs and CRC associated with calcium and vitamin D supplementation.

Similarly, the role of folate in carcinogenesis has been most studied for CRC with conflicting conclusions. Collectively, >20 case control studies have shown equivocal inverse relationships between folate status and the risk of CRC.[25] Although several large prospective studies suggest a greater reduction[25,26] in the risk of CRC and adenomas in patients with the highest intake of folate, suggesting potential benefits associated with folate supplements in mitigating the risk of CRC, other studies have shown the potential for folate supplements to increase CRC risk.[27–30]

In the Aspirin/Folate Polyp Prevention Study[27] that assessed the safety and efficacy of folic acid supplementation for preventing colorectal adenomas, folic acid did not reduce colorectal adenoma risk. Participants were randomly assigned in a 1:1 ratio to receive 1 mg/d of folic acid (n=516) or placebo (n=505), and the follow-up consisted of 2 colonoscopy surveillance cycles (the first interval at 3 years and the second at 3 or 5 years later). During the first 3 years, 987 (96.7%) participants underwent colonoscopy follow-up, and the incidence of at least 1 colorectal adenoma was 44.1% for folic acid (n=221) and 42.4% for placebo (n=206) (unadjusted RR, 1.04; 95% CI, 0.90–1.20; P=.58). The incidence of at least 1 advanced lesion was 11.4% for folic acid (n=57) and 8.6% for placebo (n=42) (unadjusted RR, 1.32; 95% CI, 0.90–1.92; P=.15). A total of 607 (59.5%) participants underwent a second follow-up, and the incidence of at least 1 colorectal adenoma was 41.9% for folic acid (n=127) and 37.2% for placebo (n=113) (unadjusted RR, 1.13; 95% CI, 0.93–1.37; P=.23), and the incidence of at least 1 advanced lesion was 11.6% for folic acid (n=35) and 6.9% for placebo (n=21) (unadjusted RR, 1.67; 95% CI, 1.00–2.80; P=.05).

During the additional follow-up after 3 years, among those who extended treatment, any colorectal neoplasia was found in 36% (n=118) participants assigned to placebo and 43% (n=146) assigned to folic acid during the second surveillance interval (RR, 1.21; 95% CI, 0.99–1.47; P=.06). Increased risk of SSL with extended folic acid supplementation was statistically significant during the second surveillance interval (RR, 1.94; 95% CI, 1.02–3.68; P=.04) and the study concluded that folic acid may increase SSL risk.[28] A few prospective, case-controlled studies have also independently demonstrated that low folate status is associated with a reduced CRC risk.[29,30]

Considering these controversies among various epidemiologic studies, in this update to the NCCN Guidelines (Version 1.2024), the panel added a cautionary statement under lifestyle/dietary factors associated with increased CRC risk stating that although supplemental calcium, vitamin D, and folate use have all been linked to a decreased risk of conventional adenomas, some evidence suggests that these agents may increase the risk of serrated polyps (Figure 1).

Aspirin

The US Preventive Services Task Force (USPSTF) conducted a systematic evidence review of trials that assessed the benefits and harms of aspirin in primary cardiovascular disease (CVD) and CRC prevention.[31] The 12 trials (including 1 pilot trial) included in this systematic review compared the effects of low-dose aspirin (≤100 mg/d) versus placebo or no treatment in adults aged ≥40 years. For events occurring within trial periods (4 trials; n=86,137), low-dose aspirin had no statistically significant association with CRC incidence at 5 to 10 years of follow-up (OR, 1.07; 95% CI, 0.92–1.24). No statistically significant beneficial association between 10 years of randomized low-dose aspirin and CRC incidence was found in any reported trials.

Based on 2 trials (the Women's Health Study and Thrombosis Prevention Trial),[32,33] aspirin use for 7 to 10 years was associated with a significantly lower risk of CRC mortality *only* when considering long-term observational follow-up (at 20 years) beyond trial periods (OR, 0.77; 95% CI, 0.61–0.98). The USPSTF recommends that the decision to initiate low-dose aspirin use for the primary prevention of CVD in adults aged 40 to 59 years who have a 10-year CVD risk of ≥10% should be individualized, and recommends against initiating low-dose aspirin use for the primary prevention of CVD in adults aged ≥60 years.[34] However, there is limited trial evidence showing benefits for CRC due to differences in duration of aspirin use, timing of outcome measurements, and length of follow-up. Therefore, the USPSTF concluded that the evidence is inadequate that low-dose aspirin use reduces CRC incidence or mortality.[34,35]

A systematic study of 11 randomized controlled trials[36] found that at 3 years, aspirin statistically reduced the risk of colorectal adenomas (RR, 0.84; $P<.05$) but not advanced lesions (RR, 0.82; $P=.10$). At 5 years, the risk of advanced lesions was significantly reduced (RR, 0.68; $P<.05$), but not in nonadvanced adenomas (RR, 0.87; $P=.22$). Beyond 5 years, aspirin had no effect on the risk of advanced lesions (hazard ratio [HR], 0.82; $P=.07$) nor adenomas (HR, 0.99; $P=.82$).[36] A similar meta-analysis reported a reduced recurrence of adenomas (RR, 0.83; 95% CI, 0.72–0.99; $P=.006$) and reduced mortality of CRC (RR, 0.79; 95% CI, 0.64–0.97; $P=.02$).[37] The ASPREE trial randomized patients aged ≥70 years to either aspirin (n=9,525) or placebo (n=9,589).[38] In contrast to the other studies, this trial reported that aspirin use was associated with a statistically significant increase in CRC mortality at 4.7 years of follow-up (HR, 1.77; 95% CI, 1.02–3.06).[38] Based on the updated USPSTF recommendation, the NCCN panel revised the aspirin section to reflect the lack of clarity on aspirin use in reducing the risk of CRC incidence or mortality when used for primary prevention (Figure 1).

To examine the secondary preventive impact of aspirin, an observational, population-based, retrospective cohort study of patients diagnosed with CRC from 2004 to 2011 in the Cancer Registry of Norway (n=23,162) examined the effect of aspirin after CRC diagnosis in 6,102 patients.[39] After a median follow-up of 3 years, the mortality rate from all causes was lower in patients who were exposed to aspirin (32.9%) compared with those who were not exposed (42.3%). In addition, aspirin exposure after CRC diagnosis was independently associated with improved CRC-specific survival (HR, 0.85; 95% CI, 0.79–0.92) and overall survival (HR, 0.95; 95% CI, 0.90–1.01).[39] A cost-effectiveness analysis also suggested that the risk/benefit profile favored the use of very-low-dose aspirin for secondary prevention in individuals with previous advanced colorectal adenomas.[40] For secondary prevention, the NCCN panel retains the statement on the association of aspirin with improved CRC-specific and overall survivals (Figure 1).

In the double-blind, randomized CAPP2 trial,[41] the study reported long-term cancer outcomes for patients with Lynch syndrome enrolled onto a randomized trial of daily aspirin (600 mg) versus placebo. The participants were followed for a mean of 10 years for a longer-term assessment and 40 (9%) of 427 participants who received aspirin developed CRC compared with 58 (13%) of 434 who received placebo (HR, 0.65; 95% CI, 0.43–0.97; $P=.035$). The ongoing CAPP3 trial, which focuses on finding the right dose of aspirin for people with a mismatch repair gene defect—the underlying cause of Lynch syndrome—has completed recruitment and the results are expected to be published in 2025. This study supported the case for aspirin use for prevention of CRC in patients with Lynch syndrome, and the pertinent statement was revised in the NCCN Guidelines (Figure 1).

Risk Assessment

The NCCN Guidelines for CRC Screening stratify patients into 2 groups depending on their risk of developing CRC: average risk and increased risk (Figures 2 and 3). Communication with the patient and referring physician regarding any updated CRC risk assessment and screening plan based on family history, colonoscopy, and pathology findings is highly encouraged.

Average Risk

According to the previous version of NCCN Guidelines for CRC Screening (Version 1.2023), individuals at average risk of developing CRC are those aged ≥45 years; with no personal history of adenoma or SSLs or CRC, inflammatory bowel disease, high-risk CRC genetic syndromes, cystic fibrosis, or childhood cancer; and with a negative family history for CRC, confirmed advanced adenoma, or an advanced SSL. At the 2024 panel meeting, the age recommendation for screening was revised to 45 to 75 years and the list of high-risk hereditary CRC syndromes was updated (Figure 2).

Approximately 43% of early-onset (age <50 years) CRC diagnoses are in individuals aged 45 to 49 years, and therefore screening in this population becomes important. Since 2010, the incidence of regional-stage and distant-stage disease has increased by approximately 3% per year in people with CRC aged <50 years. From 2011 to 2019, CRC incidence rates have increased by 1.9% per year in people aged <50 years. In addition, in 2023, there were an estimated 52,550 CRC deaths, including 3,750 decedents (7%) aged <50 years.[1]

Different CRC screening guidelines have been issued by many organizations, such as the American Cancer Society (ACS), USPSTF, American College of Gastroenterology (ACG), and American College of Physicians (ACP). At the most recent annual meeting, the NCCN panel reviewed these guidelines and other existing data in framing the age recommendation for CRC screening. A general consensus exists among the different organizations that average-risk adults aged 50 to 75 years should be screened. The recommendations vary slightly in the recommended age to initiate screening.[42] In 2018, results from modeling analyses[43] identified efficient and model-recommendable strategies that started screening at age 45 years, and ACS recommended that screening begin at age 45 years in all adults (qualified recommendation).[43] In 2021, the microsimulation modeling analysis[44] performed to inform USPSTF on CRC screening strategies suggested that CRC screening starting at age 45 years provided an efficient balance of colonoscopy burden and life-years gained. Based on this analysis, the USPSTF recommended screening for CRC in adults aged 45 to 49 years,[45] and the ACG also suggested screening in average-risk persons aged 45 to 49 years as a conditional recommendation.[4]

Regarding the stopping age of CRC screening, generally these guidelines agree that screening should be either individualized in older adults aged 76 to 85 years (ACS, American Academy of Family Physicians, and US Multi-Society Task Force) or

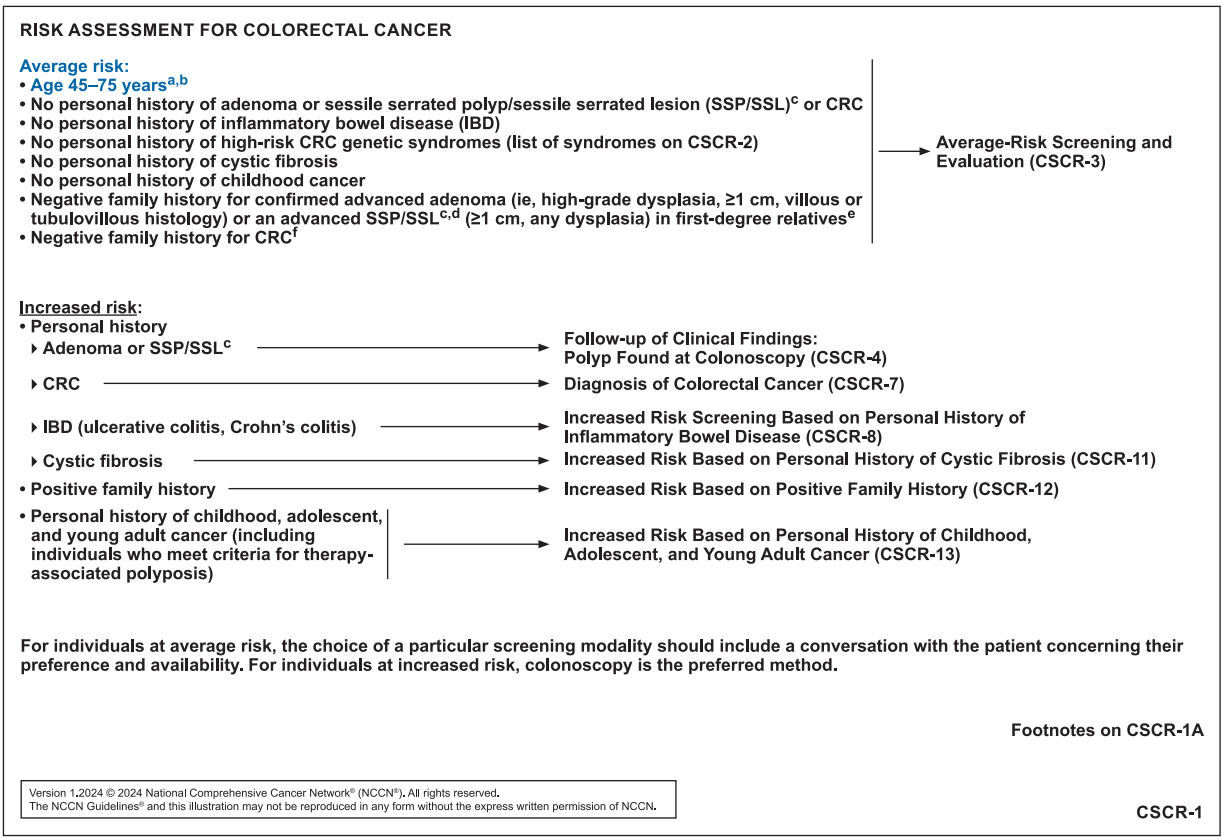


Figure 2. CSCR-1. NCCN Clinical Practice Guidelines in Oncology for Colorectal Cancer Screening, Version 1.2024.

stopped altogether (ACP), with clear consensus that screening should stop after 85 years of age[45] (Figure 2).

Based on their discussion, the NCCN panel re-emphasized a footnote that states that CRC screening is recommended in adults aged 45 to 75 years who might have a life expectancy of ≥10 years. The decision to screen between ages 76 and 85 years should be individualized and include a discussion of the risks and benefits based on comorbidity status and estimated life expectancy. In this age group, eligible individuals who have not been previously screened are the most likely to experience benefit.

Although age consideration may be dependent on race/ethnicity, patient preference, and resources available, prompt evaluation of alarm symptoms in patients aged <45 years is of critical importance. Epidemiologic reports suggest that CRC incidence is increasing in young adults, with nearly half of patients presenting with early-onset CRC being <45 years of age for unknown reasons.[1,9,46] Many of these young adults have signs and symptoms of CRC, such as iron deficiency, anemia, rectal bleeding, or a change in bowel habits.

To date, patients aged <45 years with early-onset CRC are prone to experience diagnostic delays and missed diagnostic opportunities[47,48] due to lack of awareness of red-flag signs and symptoms. Because routine testing is not recommended for patients aged <45 years, symptom recognition remains an important component of early detection and treatment in this patient population.

Numerous studies have suggested that anemia, stomach pain, and constipation are the most common symptoms of the group with missed diagnostic opportunities for CRC.[48–51] In a matched, case-control study[52] utilizing longitudinal claims data with 5,075 early-onset CRC cases, 4 red-flag signs and symptoms (abdominal pain, rectal bleeding, diarrhea, and iron deficiency anemia) between 3 months and 2 years prior to the index date were associated with increased risk of early-onset CRC, and the strongest association was found with rectal bleeding. The presence of 1 of these symptoms was associated with a 1.9-fold increased risk, 2 of the symptoms was associated with a 3.6-fold increased risk, and at least 3 of these red-flag signs and symptoms were associated with a 6.5-fold increased risk. The panel discussed these studies and revised the section on the evaluation of alarm symptoms in patients aged <45 years (Figure 3).

Increased Risk Based on Personal History of Childhood, Adolescent, and Young Adult Cancer

Therapy-associated polyposis (TAP) is an acquired phenotype in childhood cancer survivors that presents years after exposure to chemotherapy and/or radiation therapy.[53] If an individual has a cumulative incidence of ≥10 gastrointestinal polyps of any type (including adenoma, SSLs, or hamartomas) in the entire gastrointestinal tract, a history of systemic therapy and/or radiotherapy for a childhood or young adult cancer (specifically abdominopelvic radiotherapy and/or alkylating chemotherapy), and completed multigene testing without an identified pathogenic variant, then a baseline upper endoscopy is indicated if colonic polyposis is identified (Figure 4). Multigene testing should include all hereditary polyposis and CRC genes.[54] Germline multigene panel testing has

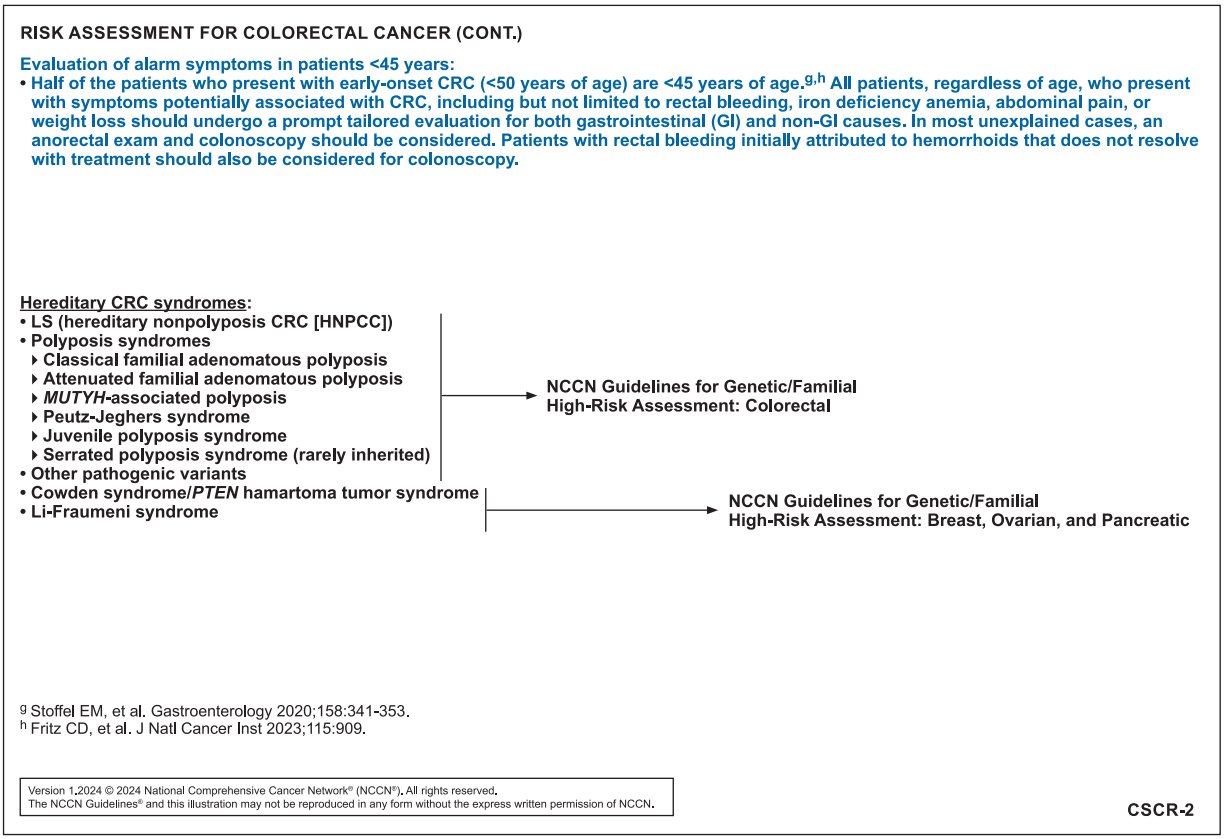


Figure 3. CSCR-2. NCCN Clinical Practice Guidelines in Oncology for Colorectal Cancer Screening, Version 1.2024.

been updated to include at minimum the following CRC risk–associated genes: *APC*, *MUTYH*, *MLH1*, *MSH2*, *MSH6*, *PMS2*, *EPCAM*, *BMPR1A*, *SMAD4*, *PTEN*, *STK11*, and *TP53*. Pathogenic variants associated with adenomatous polyposis include, but are not limited to, monoallelic pathogenic variants in *APC*, *GREM1*, *POLE*, *POLD1*, and *AXIN2*, and biallelic pathogenic variants in *MUTYH*, *NTHL1*, and *MSH3*. Patients with TAP have phenotypic features that resemble numerous hereditary CRC syndromes, suggesting multiple concurrent biologic mechanisms. Recognition of TAP diagnosis is important for cancer risk assessment and screening for childhood and young adulthood cancer (CYAC).

In 2022, a new algorithm page was added with surveillance modality and schedule for increased-risk individuals based on personal history of childhood, adolescent, and young adult cancer. CYAC survivors who received prior radiation therapy (particularly to the abdominopelvic field) or total body irradiation (regardless of dose, with or without chemotherapy) are at an increased risk for CRC. The Children's Oncology Group (COG) long-term follow-up guidelines[55] recommend colonoscopy starting at 30 years of age or 5 years after treatment (whichever occurs last) and repeating every 5 years, and the NCCN panel added this to the recommendation. For CYAC survivors with a history of chemotherapy only (without radiation), based on a multi-institutional cohort study of TAP in CYAC by Biller et al,[56] the previous recommendation was to initiate colonoscopy screening at 35 years or 10 years after chemotherapy treatment (whichever occurs first) and repeat every 5 years.

In that TAP cohort study,[56] 34 patients were included; 29 (85%) received chemotherapy for their initial CYAC and 28 (82%) received radiation therapy. At the time of their original diagnosis, the median age of the CYAC survivors was 18 years. Gastrointestinal polyposis was first detected at a median of 27 years after CYAC treatment. Among 34 patients with TAP, 12 (35%) had ≥50 colorectal polyps, 32 (94%) had >1 histologic polyp type, and 25 (74%) had clinical features suggestive of ≥1 CRC predisposition syndrome, including 8 (24%) with features of multiple such syndromes. In this cohort, almost 20% had polyps first detected at an age prior to the COG recommended colonoscopy screening start-time (ie, at 30 years of age or 5 years after radiation, whichever occurs last), and patients with and without prior radiation therapy fell outside of the COG screening guidelines.

Based on this TAP cohort, Biller et al[56] proposed that the COG guidelines be expanded to include CYAC survivors who received chemotherapy (without radiation), and that screening initiation begin at 35 years of age or 10 years after chemotherapy, whichever occurs first. On closer look at the TAP cohort, the median age of the CYAC survivors who received only chemotherapy was 22 years and the recommendation to initiate screening at whichever age recommendation occurs first is applicable only to this cohort. To expand the applicability of this guideline to younger CYAC survivors, multiple NCCN Member Institutions requested a review of the screening recommendation in CYAC survivors with a history of chemotherapy at the annual panel meeting.

The NCCN panel discussed extensively the surveillance modality and schedule for CYAC survivors with a personal history of chemotherapy (without radiation therapy) and updated the recommendation to initiate colonoscopy screening beginning at

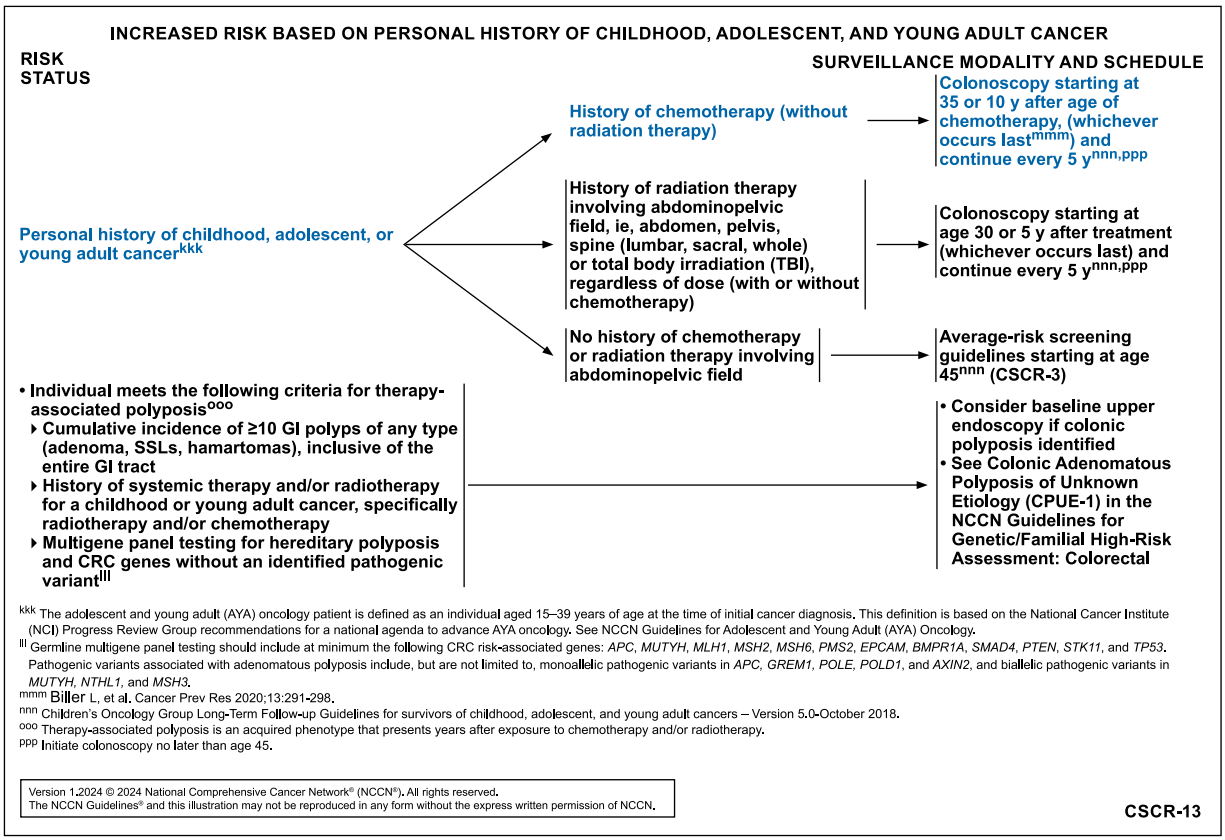


Figure 4. CSCR-13. NCCN Clinical Practice Guidelines in Oncology for Colorectal Cancer Screening, Version 1.2024.

35 years of age or 10 years after chemotherapy treatment, whichever occurs last, and repeat every 5 years (Figure 4).

Conclusions

Several factors are important and should be considered for CRC screening, including age of initiation and ending; consideration of dietary factors, including dietary supplements and aspirin use; and risk assessment of CYAC survivors. During the recent update of the NCCN Guidelines for CRC Screening, the panel addressed several issues, including those discussed herein, and clarified several important recommendations in the NCCN Guidelines. Recently published and ongoing studies were also discussed and emerging data will continue to inform the panel's recommendations.

To participate in this journal CE activity, go to https://education.nccn.org/node/94867

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